SOFTWARE ENGINEERING — LAB 6

Code and Branch Coverage Analysis — Food Delivery

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1. Aim and objectives

The aim of this lab is to measure and improve the test coverage of the assigned Food Delivery Python application. The work compares statement (line) coverage with branch coverage and uses targeted tests to exercise normal cases, validation failures, coupon rules, distance bands, weather surcharge, and delivery-time decisions.

2. Assigned application

Repository: https://github.com/SETAPESU26/L6_L7_20_food_delivery

Submission repository: https://github.com/A6h1ramahy/PES2UG24CS023_SE-Lab/tree/main/LAB6

The application has three functions: delivery_charge(distance_km, order_value, is_raining), apply_coupon(order_value, code, is_first_order), and estimated_time(distance_km, is_peak_hour, restaurant_busy). The original application logic is preserved in food_delivery.py.

3. Test strategy and coverage criteria

3.1 Code / line coverage stage

Line coverage checks whether executable statements in the application were executed at least once. The first test file deliberately exercises every executable statement, including each return statement, exception path, fee calculation, coupon deduction, and delivery-time adjustment.

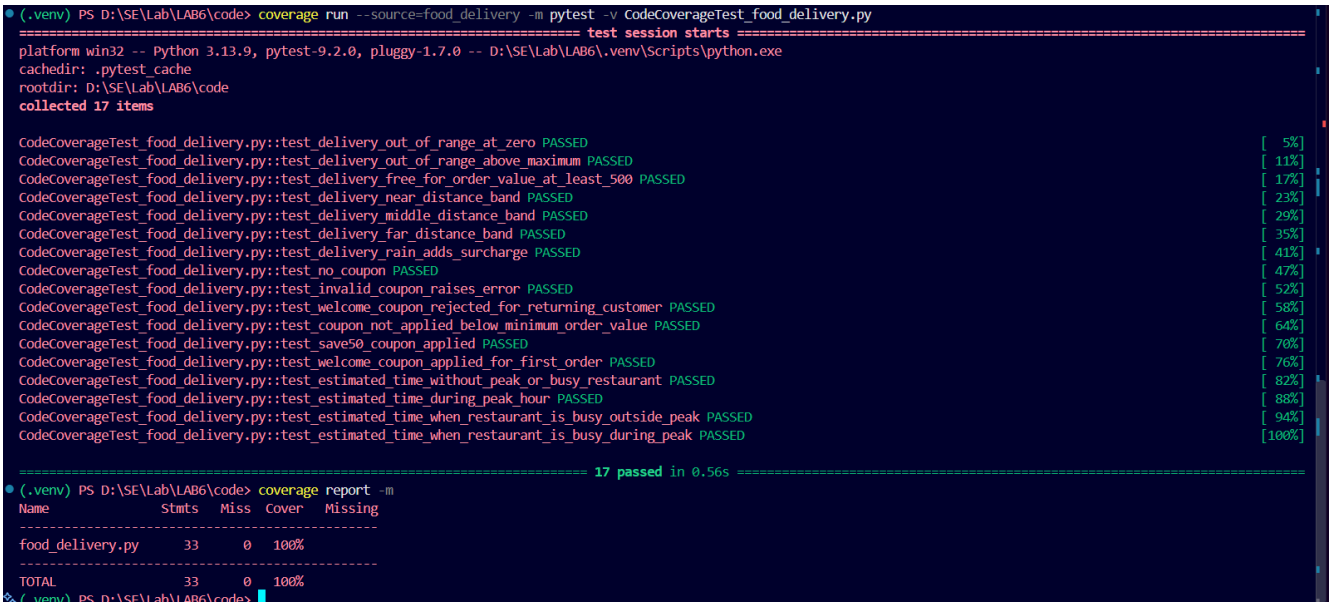
3.2 Branch coverage stage

Branch coverage checks whether each decision outcome has been exercised. Tests therefore cover both sides of the delivery-range check, the free-order check and distance tiers, raining and non-raining cases, coupon absent/invalid/valid cases, first-order eligibility, the minimum order threshold, peak/non-peak time, busy/not-busy restaurant, and the nested busy-plus-peak decision.

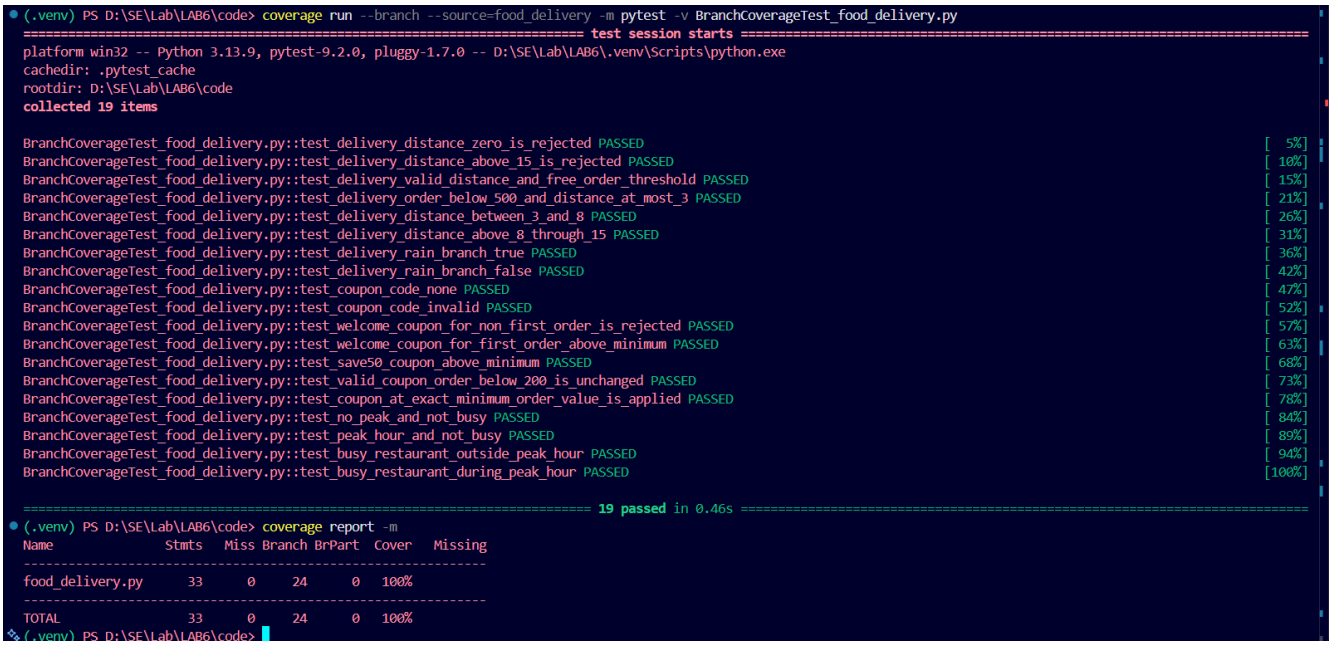
Coverage is a test adequacy measure, not proof that the application is defect-free. The tests use the expected behavior defined by the assigned source code.

4. Screenshots

4.1 Code coverage stage



4.2 Branch coverage stage



6. Test case coverage matrix

Function	Decision / condition	Expected behavior covered
delivery_charge	distance <= 0 / distance > 15	Invalid range raises ValueError
delivery_charge	order_value >= 500	Free delivery charge
delivery_charge	distance <= 3	Near-distance fee
delivery_charge	3 < distance <= 8	Middle-distance fee
delivery_charge	distance > 8 and <= 15	Far-distance fee
delivery_charge	is_raining True / False	Rain surcharge and no surcharge
apply_coupon	code is None	Original order value returned
apply_coupon	unknown coupon	Invalid coupon ValueError
apply_coupon	WELCOME and not first order	Eligibility ValueError
apply_coupon	order value < 200	Coupon not applied
apply_coupon	order value >= 200 and valid coupon	SAVE50 / WELCOME discount applied
estimated_time	peak hour True / False	Peak adjustment and no peak adjustment
estimated_time	restaurant_busy True / False	Busy adjustment and no busy adjustment
estimated_time	busy and peak nested condition	Additional nested 5-minute adjustment

THE END